Athena Test Document

This document is created for testing the Athena offline RAG pipeline.

Purpose

Athena is an offline artificial intelligence research assistant designed to index personal documents, search relevant passages, and generate answers using local language models.

Architecture

The Athena system contains several stages:
1. Document extraction converts files into searchable text.
2. Chunking divides documents into smaller passages.
3. Embedding generation converts text into numerical vectors.
4. Semantic retrieval finds relevant passages.
5. Prompt building prepares context for the language model.
6. Ollama provides local AI generation.

Research Workflow

A user imports a document into Athena. The document is processed locally. Athena stores document chunks and embeddings. When a question is asked, the system retrieves relevant passages and produces an answer with source references.

Example Knowledge

Athena was designed to operate without cloud dependency. Local models such as Gemma, Mistral, and DeepSeek can be used for answering questions. Document privacy is maintained because files remain on the user's computer.

Test Questions

1. What is Athena?
2. How does Athena process documents?
3. Which models can Athena use?
4. Why is local AI useful for document research?

Expected Answer

Athena is an offline AI research assistant that combines document indexing, semantic search, retrieval augmented generation, and local language models.